

Technical Document #930: Retrieval Augmented Generation - Comprehensive Overview

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Chunking splits documents into smaller segments for embedding. Optimal chunk size balances context retention with retrieval precision.

Hybrid search combines keyword-based and semantic search. It leverages the strengths of both approaches for better retrieval quality.

Context window optimization selects the most relevant information to fit within the LLM's context limit. It maximizes information density.

Computer vision applications are becoming ubiquitous, from facial recognition to autonomous driving. Deep learning has enabled breakthroughs in visual understanding.

Cybersecurity threats continue to evolve in complexity and scale. Organizations must invest in robust security measures to protect sensitive data.

The Internet of Things (IoT) connects billions of devices worldwide. This connectivity generates massive amounts of data that can be analyzed for insights.

Natural language processing has made significant advances with large language models. These models can understand and generate human-like text with remarkable fluency.

Augmented reality overlays digital information on the physical world. It has applications in training, maintenance, and entertainment.

Bioinformatics applies computational methods to analyze biological data. It has accelerated drug discovery and our understanding of genetic diseases.

Key Takeaways:

- Hallucination detection identifies when LLMs generate factually incorrect information.
- Query decomposition breaks complex queries into simpler sub-queries.
- Hybrid search combines keyword-based and semantic search.